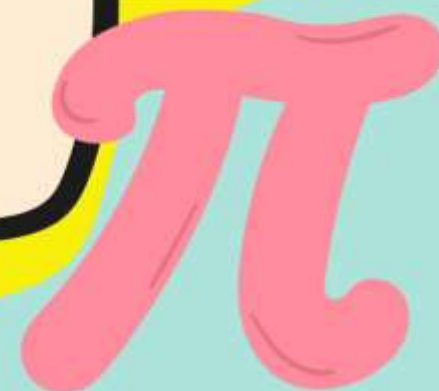




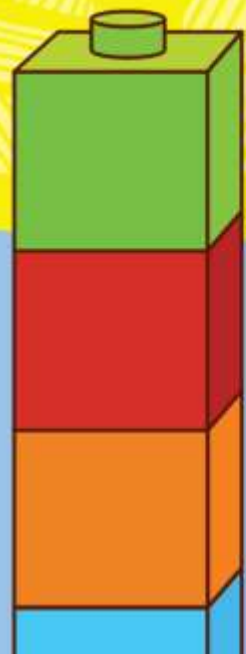
STORM

STRIVE "A" FOR MATHS

SM015

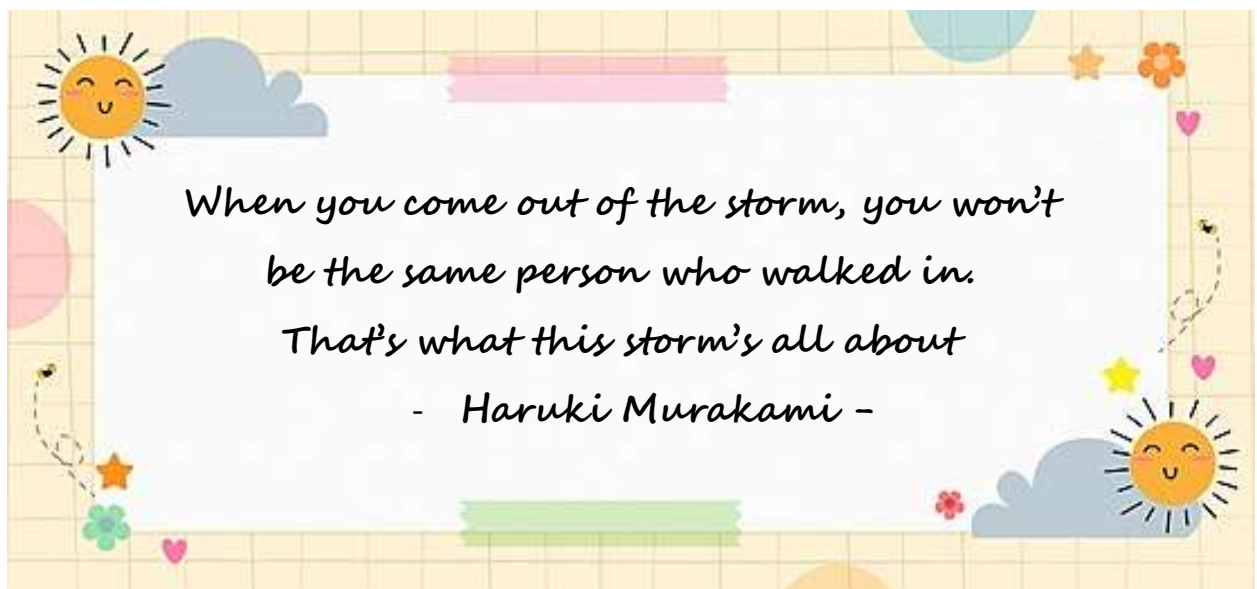


NAME:
CLASS:
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Mohamad Azhar Bin Hassan
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LIST OF MATHEMATICAL FORMULAE

Trigonometry ***

$$\sin (A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\begin{aligned}\cos 2A &= \cos^2 A - \sin^2 A \\ &= 2 \cos^2 A - 1 \\ &= 1 - 2 \sin^2 A\end{aligned}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

Differentiation

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{d^2 y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

Sphere
Sfera

$$V = \frac{4}{3} \pi r^3$$

$$S = 4 \pi r^2$$

Right circular cone
Kon membulat tegak

$$V = \frac{1}{3} \pi r^2 h$$

$$S = \pi r^2 + \pi r h$$

Right circular cylinder
Silinder membulat tegak

$$V = \pi r^2 h$$

$$S = 2\pi r^2 + 2\pi r h$$

*** Subject to change

SET 1

TIME: 2 HOURS

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** the questions given.

1. Solve

(a) $\lim_{x \rightarrow -3} \frac{x^3 + 27}{x + 3}$

(b) $\lim_{x \rightarrow 2} \frac{\sqrt{x^2 + 12} - 4}{x - 2}$

[9 marks]

2. By expressing $y = \frac{1}{(x-1)^2(x-3)}$ into partial fraction, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

Give your answer in the simplest form. Hence, evaluate $\frac{d^2y}{dx^2}$ when $x = -1$.

[9 marks]

3. Find the critical numbers for $f(x) = x^4 - 8x^3 + 18x^2 - 27$. Hence, determine the points of inflection.

[7 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answers **all** the questions given.

1. Given $z_1 = 6 + 3i$ and $z_2 = 5 - 2i$.

(a) Verify $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$.

(b) Find $\text{Arg}(z_1)$.

[5 marks]

2. Solve

(a) $x^{\ln x - 2} = e^3.$

(b) $|2x + 3| > |x - 5|, x \neq 5$

[11 marks]

3. Given $f(x) = 4^{x+2}.$

(a) Find the value of a when $f(a) = 1.$

(b) Find $f^{-1}(x)$ and its domain.

[7 marks]

4. Given $f(x) = e^{-2x} + 3; (g \circ f)(x) = \frac{1}{2} \ln(e^{-2x} + 2).$

(a) Find domain and range of $f(x).$

(b) Find $g(x)$

(c) Hence, find $g^{-1}(x).$ Sketch the graph of $g(x)$ and $g^{-1}(x)$ on the same axes.

[13 marks]

5 (a) Evaluate $\lim_{x \rightarrow -\infty} \frac{3x + 2}{\sqrt{4x^2 - 3}}.$

(b) Given $f(x) = \begin{cases} ax + 3, & x > 5, \\ 8, & x = 5, \\ x^2 + bx - 2a, & x < 5. \end{cases}$

Find a and b such that the function is continuous for all $x.$

(c) Given $y = \frac{x+1}{(x-4)(x+2)}.$

By finding the limit at appropriate points, obtain the horizontal and vertical asymptotes.

[14 marks]

- 6 (a) Given $y^3 = e^{x-y}$. Show that.

(i) $\frac{dy}{dx} = \frac{y}{3+y}$

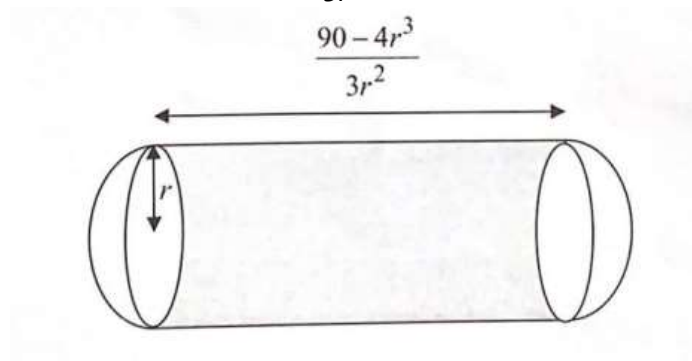
(ii) $(3+y)^3 \frac{d^2y}{dx^2} - 3y = 0$

- (b) Given the parametric equations for the curve
 $x = 2 + \sqrt{\cos 2t}$ and $y = 1 - \sqrt{\sin 2t}$.

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. Give your answer in the simplest form.

[16 marks]

7. The figure below shows a cylinder with both ends are two hemispheres with radius, r meter and length $\frac{90-4r^3}{3r^2}$ meter.



- (a) Show that the total surface area is $A = \frac{4}{3}\pi r^2 + \frac{60\pi}{r}$.
- (b) Hence, find the value of r when the total surface area is minimum.

[9 marks]

END OF QUESTION PAPER

Final Answer**SECTION A**

1. (a) 27
(b) $\frac{1}{2}$
2. (a) $y = \frac{1}{(x-1)^2(x-3)} = -\frac{1}{4(x-1)} - \frac{1}{2(x-1)^2} + \frac{1}{4(x-3)}$
 $\frac{dy}{dx} = \frac{1}{4(x-1)^2} + \frac{1}{(x-1)^3} - \frac{1}{4(x-3)^2}$
 $\frac{d^2y}{dx^2} = -\frac{1}{2(x-1)^3} - \frac{3}{(x-1)^4} + \frac{1}{2(x-3)^3}$
 $\left. \frac{d^2y}{dx^2} \right|_{x=-1} = -\frac{17}{128}$
3. Critical number: $x = 0$ and $x = 3$, Point of inflection : $(1, -16)$ and $(3, 0)$

SECTION B

1. (a) Shown
(b) 0.464 rad
2. (a) $x = e^3$ $x = e^{-1}$
(b) $(-\infty, -8) \cup \left(\frac{2}{3}, 5\right) \cup (5, \infty)$
3. (a) -2
(b) $f^{-1}(x) = \log_4 x - 2$, $D_{f^{-1}} = (0, \infty)$
4. (a) $D_f = (-\infty, \infty)$, $R_f = (3, \infty)$
(b) $g(x) = \frac{1}{2} \ln(x-1)$
(c) $g^{-1}(x) = e^{2x} + 1$
5. (a) $-\frac{3}{2}$
(b) $a = 1$, $b = -3$
(c) Horizontal asymptote $y = 0$
Vertical asymptote $x = 4$, $x = -2$

6. (a) (i) Shown
(ii) Shown
(b) $\frac{dy}{dx} = \cot^{\frac{3}{2}} 2t$, $\frac{d^2y}{dx^2} = 3 \cos 2t \operatorname{cosec}^{\frac{7}{2}} 2t$
7. (a) Shown
(b) $r = 2.823$

SET 2**TIME: 2 HOURS****SECTION A [25 marks]**

This section consists of **3** questions. Answers **all** the questions given.

1. (a) Given the function

$$f(x) = \begin{cases} x+1 & , x < 1 \\ \frac{x^2 - x - a}{x - a} & , x \geq 1, x \neq a \end{cases}$$

is continuous at $x = 1$. Find the value of a . Hence, determine

$$\lim_{x \rightarrow a} f(x).$$

[6 marks]

(b) Evaluate $\lim_{x \rightarrow +\infty} \left(\frac{2x+5}{3x-4} \right)^{\frac{1}{3}}.$

[3 marks]

2. (a) Given the parametric equation

$$x = \frac{1}{t} + 3t^2 \quad y = t^2 - 4\sqrt{t}, \quad t > 0.$$

Find $\frac{d^2y}{dx^2}$ in terms of t . Hence, evaluate $\frac{d^2y}{dx^2}$ at $t = 1$.

[6 marks]

(b) Given $y = \ln \left(\frac{\sin 2x}{1 - \cos 2x} \right)$. Show that $\frac{dy}{dx} = -2\operatorname{cosec} 2x$.

[5 marks]

3. A farmer has a 100 meter of fencing wants to fence off a rectangular piece of land that borders a straight river. The part of the land which borders the river needs no fencing. Find the dimensions of the enclosed that has the maximum area.

[5 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answers **all** the questions given.

1. (a) Solve the equation

$$\ln 2^{4x-1} = \ln 8^{x+5} + \log_2 16^{1-2x}$$

by expressing your answer in terms of $\ln 2$.

[4 marks]

- (b) Given $z_1 = 2 - 3i$ and $z_2 = 4 + 2i$.

Without using calculator, find $z_3 = \frac{\bar{z}_1}{z_2}$ and hence, evaluate $|z_3|$.

[5 marks]

2. Determine the solution set of the following inequalities.

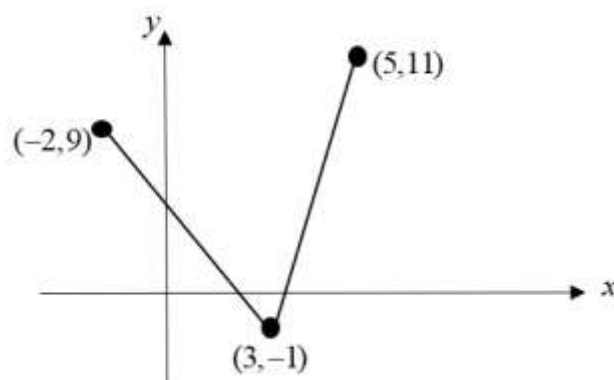
(a) $\frac{4x}{4-x} \geq 6$.

[5 marks]

(b) $|x-6| < 3|x-2|$.

[6 marks]

3. FIGURE 1 shows a function $f(x)$ with domain $-2 \leq x \leq 5$.



- (a) State the range of $f(x)$.

[1 marks]

- (b) Determine the function of $f(x)$.

[4 marks]

- (c) Evaluate $(f \circ f)(0)$.

[2 marks]

4. Given the function $f(x) = \frac{3}{x^2 - 4}, x \neq -2, 2$ and $(f \circ g)(x) = \frac{3}{x - 5}, x \neq 5$.

(a) Obtain $g(x) \geq 0$ and state its domain.

[4 marks]

(b) Show that $g^{-1}(x)$ exists. Hence, find $g^{-1}(x)$ and its domain.

[5 marks]

(c) Find the values a and b if $h(x) = ax + b$, such that

$$(h \circ g^{-1})(x) = 5x^2 + 12.$$

[4 marks]

5. Given $f(x) = \frac{2x^2 + 2}{x^2 - 4}$.

(a) Find the vertical and horizontal asymptotes of $f(x)$.

[8 marks]

(b) Hence, sketch the graph $f(x)$.

[3 marks]

6. (a) Given $y = \tan\left(\frac{\pi}{4} + x\right)$, show that $\frac{d^2y}{dx^2} - 2y\frac{dy}{dx} = 0$.

[5 marks]

(b) Differentiate $g(x) = \frac{e^x}{e^x - 2}, x \neq \ln 2$ with respect to x .

Hence, calculate the exact value of x if $g'(x) = -2$.

[6 marks]

(c) Find the value of tangent to the curve $y^2 = x^3$ and $2x^2 + 3y^2 = k$, where k is a constant, at the points $(1, -1)$ and $(1, 1)$.

Hence, state the relationship between the tangent values at these two points.

[5 marks]

7. Consider a cone as shown in FIGURE 2.

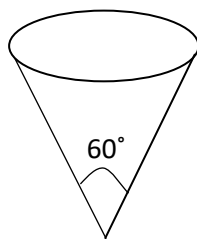


FIGURE 2

Water flows out from a cone through a hole located at the bottom tip at the rate of $5\text{cm}^3/\text{s}$. Find the rate of change of the height of the water when the volume of water in the cone is $3000\pi\text{cm}^3$.

[8 marks]

END OF QUESTION PAPER

Final Answer:

SECTION A

1. (a) $a = 2$, $\lim_{x \rightarrow 2} f(x) = 3$
 (b) $(2/3)^{\frac{1}{3}}$
2. (a)
$$\frac{\left(-\frac{1}{t^2} + 6t\right)\left(2 - 2\left(-\frac{1}{2}\right)t^{-\frac{3}{2}}\right) - \left(2t - \frac{2}{\sqrt{t}}\right)\left(\frac{2}{t^3} + 6\right)}{\left(-\frac{1}{t^2} + 6t\right)^3}, \frac{3}{25}$$

 (b) Show
3. $x = 25\text{cm}$, $y = 50\text{cm}$

SECTION B

1. (a) $\frac{4 + 16\ln 2}{8 + \ln 2}$
 (b) $\frac{7}{10} + \frac{2}{5}i$, $\frac{\sqrt{65}}{10}$

2. (a) $\left\{x : \frac{12}{5} \leq x < 4\right\}$

(b) $\{x : x < 0 \cup x > 3\}$

3. (a) $[-1, 11]$

(b) $f(x) = \begin{cases} -2x+5 & , -2 \leq x \leq 3 \\ 6x-19 & , 3 \leq x \leq 5 \end{cases}$

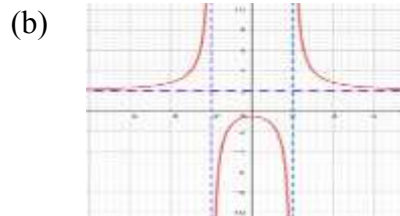
(c) 11

4. (a) $g(x) = \sqrt{x-1}$, $D_{g(x)} = x \geq 1$ / $x = 5$

(b) Show, $g^{-1}(x) = x^2 + 1$, $D_{g^{-1}(x)} = x \geq 0$ / $x = 2$

(c) $a = 5$, $b = 7$

5. (a) $x = -2$ and 2 is the vertical asymptote.
 $y = 2$ is the horizontal asymptote.



6. (a) Show

(b) $\frac{-2e^x}{(e^x - 2)^2}$, $x = 0$, $\ln 4$

(c) $-\frac{3}{2}$, $\frac{3}{2}$, $\frac{2}{3}$, $-\frac{2}{3}$, perpendicular/normal/reciprocal to each other

7. $-\frac{1}{60\pi} \text{ cm/s}$

SET 3**TIME: 2 HOURS****SECTION A [25 marks]**

This section consists of **3** questions. Answer **all** the questions given.

1. (a) Determine $\lim_{x \rightarrow \infty} \frac{5x^3 + 4x - 9}{7 - 3x^3}$.

[4 marks]

(b) The function $g(x)$ is defined by

$$g(x) = \begin{cases} m^2 - 8 & , x \leq 2, \\ \frac{x-2}{\sqrt{2x}-2} & , 2 < x \leq 8 \end{cases}$$

where m is a constant. If $\lim_{x \rightarrow 2} g(x)$ exists, find the values of m .

[5 marks]

2. (a) Find the derivative of $f(x) = \frac{1}{x+3}$ by using the first principle.

[4 marks]

(b) Given $y = x^2 e^{-3x}$, find $\frac{d^2 y}{dx^2}$. Give your answer in the simplest form.

[5 marks]

3. Given $f(x) = x^2 - \frac{6}{x}$, where $x \neq 0$. Find the coordinate(s) of stationary point

and determine whether it is a relative maximum or a relative minimum. Give the coordinates correct to 3 decimal places.

[7 marks]

SECTION B [75 marks]

This section consists of **7** questions. Answer **all** the questions given.

1. Given the complex numbers $z_1 = -i$ and $z_2 = 2 + i\sqrt{3}$, find $W = \frac{z_1^2 + \bar{z}_2}{z_1}$.

Hence, find $|W|$ and $\arg W$.

[5 marks]

2. Solve the following:

(a) $\log_2 2x = 2 \log_4 (x + 4).$

[5 marks]

(b) $2 \left| \frac{x-3}{2x-1} \right| \geq 1.$

[6 marks]

3. Given $f(x) = \frac{1}{1 + \frac{1}{1 + \frac{1}{x}}}$

(a) Simplify $f(x)$ and hence, evaluate $f\left(\frac{1}{2}\right).$

[4 marks]

(b) The domain of $f(x)$ is a set of real number except three numbers. [3 marks]
Determine the numbers.

4. Given a function $f(x) = \ln(2x+1).$

(a) State the domain and range of $f(x).$

[2 marks]

(b) Find the inverse function of $f(x)$ and state its domain and range.

Hence, find the value of x for which $f^{-1}(x) = 0.$

[7 marks]

(c) Sketch the graphs of $f(x)$ and $f^{-1}(x)$ on the same coordinate axes.

[4 marks]

5. (a) The function $g(x)$ is defined by

$$g(x) = \begin{cases} 3 & , 2 < x \leq 8 \\ \frac{|8-x|}{x-8} + k & , x > 8. \end{cases}$$

Find the values of k such that $g(x)$ is discontinuous at $x = 8.$

[6 marks]

- (b) A function $f(x)$ is defined as

$$f(x) = \begin{cases} \frac{9-x^2}{x-3} & , x < 3, \\ -6 & , x = 3, \\ \frac{162x-54x^2}{x^3-27} & , x > 3. \end{cases}$$

Determine whether $f(x)$ is continuous at $x = 3$.

[8 marks]

6. (a) Given $x^2y = \sin(2x^2 + \pi)$.

Show that $2y(1+8x^4) + 4x \frac{dy}{dx} + x^2 \frac{d^2y}{dx^2} = 4\cos(2x^2 + \pi)$.

[7 marks]

- (b) Consider the parametric equations

$$x = 2t - t^{-1}, \quad y = 2t + t^{-1}$$

- (i) Show that $\frac{dy}{dx} = \frac{2t^2-1}{2t^2+1}$.

[4 marks]

- (ii) Find $\frac{d^2y}{dx^2}$ in term of t . Hence, show that $\frac{d^2y}{dx^2} = \frac{8}{y^3}$.

[5 marks]

7. Given a right circular cylinder with height $2h$ and radius $r = \sqrt{a^2 - h^2}$, which is inscribed in a sphere of fixed radius a .

- (a) Show that the volume of the cylinder is $V = 2\pi(a^2 - h^2)h$. Hence, find the first and second derivative of the volume, V .

[5 marks]

- (b) From part 7(a), find the maximum value of the volume, V in terms of π and a as the height, h varies.

[4 marks]

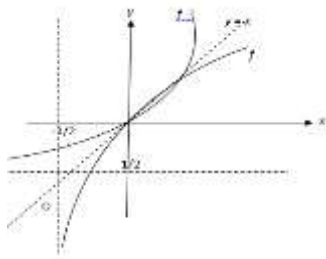
END OF QUESTION PAPER

Final Answer**SECTION A**

1. (a) $-\frac{5}{3}$
(b) $\pm\sqrt{10}$
2. (a) $-\frac{1}{(x+3)^2}$
(b) $e^{-3x}(9x^2 - 12x + 2)$
3. $(-1.442, 6.240)$ is the minimum point

SECTION B

1. (a) $z_1^2 = -1, \bar{z}_2 = 2 - i\sqrt{3}$
(b) $W = \sqrt{3} + i, |W| = 2, \arg(W) = \frac{\pi}{6} @ 0.524$
2. (a) $x = 4$
(b) $\left(-\infty, \frac{1}{2}\right) \cup \left(\frac{1}{2}, \frac{7}{4}\right]$
3. (a) $\frac{x+1}{2x+1}, \frac{3}{4}$
(b) $-1, -\frac{1}{2}, 0$
4. (a) $D_f = \left(-\frac{1}{2}, \infty\right), R_f = (-\infty, \infty)$
(b) $f^{-1}(x) = \frac{e^x - 1}{2}, D_{f^{-1}} = (-\infty, \infty), R_{f^{-1}} = \left(-\frac{1}{2}, \infty\right), x = 0$
(c)



5. (a) $k \neq 2$

(b) $f(x)$ is continuous at $x = 3$

6. (a) Shown

(b) (i) Shown

(ii)
$$\frac{8t^3}{(2t^2 + 1)^3}$$

7. (a)
$$\frac{dV}{dh} = 2\pi a^2 - 6\pi h^2 \quad , \quad \frac{d^2V}{dh^2} = -12\pi h$$

(b)
$$\frac{4\sqrt{3}\pi a^3}{9}$$

SET 4

TIME: 2 HOURS

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** the questions given.

1. The function $g(x)$ is defined by

$$g(x) = \begin{cases} 2 & , x \leq 2 \\ \frac{x-2}{\sqrt{2x}-2} & , 2 < x \leq 8 \\ \frac{|8-x|}{x-8} & x > 8 \end{cases}$$

Find

(a) $\lim_{x \rightarrow 2^+} g(x)$ [5 marks]

(b) $\lim_{x \rightarrow 8^+} g(x)$ [4 marks]

2. The parametric equations of a curve are $x = t + \frac{2}{t}$ and $y = 2t - \frac{4}{t}$ where

$t \neq 0$. Show that $\frac{dy}{dx} = 2 + \frac{8}{t^2 - 2}$. Hence, find $\frac{d^2y}{dx^2}$ in term of t .

[9 marks]

3. Given $f(x) = \frac{3x}{x^2 + 9}$ where $x > 0$. Find the coordinates of the stationary point and state its nature. [7 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** the questions given.

1. Given $z_1 = 2 + 3i$ and $z_2 = 4 - 4i$. Express $\frac{z_2}{z_1} + \left[\frac{i^3}{-z_2} \right]$ in Cartesian form.

[5 marks]

2. (a) Solve $\sqrt{6x+1} - \sqrt{x} = 3$ [5 marks]

- (b) Determine the solution set of x which satisfies the inequality

$$\frac{2}{x+1} < \frac{x}{x+3}$$
 [6 marks]

3. Given $f(x) = 2x + 1$ and $g(x) = \frac{-3x}{x+1}$.

- (a) Obtain $(f \circ g)(x)$ and $(g \circ f)(x)$. [3 marks]

- (b) Hence, determine the value of constant k where $(f \circ g)(k) = (g \circ f)(k)$ [4 marks]

4. (a) Given $f(x) = \left(\frac{5x+1}{4x}\right)$ and $g(x) = \frac{\sqrt{x+1}-2}{x^2-4}$. Find

(i) the domain of $g(x)$.

(ii) $h(x)$, if $(f \circ h)(x) = x$. [6 marks]

- (b) Given $p(x) = \ln 3x + 6$ and $q(x) = \frac{e^x}{3} - 2$. Show that $p(x)$ and $q(x)$ are inverses of each other. [7 marks]

5. (a) Evaluate $\lim_{x \rightarrow 4} \left(\frac{4-x}{2-\sqrt{x}} \right)$. [3 marks]

- (b) The function $f(x)$ is defined by

$$f(x) = \begin{cases} x-4 & x < 2 \\ A & x = 2 \\ 2x+B & 2 < x \leq 5 \\ C & x > 5 \end{cases}$$

where A, B and C are constants.

- (i) Find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$. [3 marks]

- (ii) Use the definition of continuity to determine the values of A and B if $f(x)$ is continuous at $x = 2$. [4 marks]

- (iii) Show that if $\lim_{x \rightarrow 5} f(x)$ doesn't exist, then $C \neq 4$. Is $f(x)$ continuous at $x = 5$ when $C \neq 4$? [4 marks]

6. (a) Find the derivative of $f(x) = \frac{6}{\sqrt{x}}$ using the first principle. [8 marks]
- (b) Find the value of $\frac{dy}{dx}$ when $x = 0$ for each of the following:
- (i) $y = \ln 9 - 2x$
- (ii) $y = \frac{e^{-3x}}{\sqrt{3x+1}}$ [8 marks]
7. Water is poured into a right inverted cone of height h with a semi vertical angle of 60° at a constant rate of $25\pi \text{ cm}^3$ per second.
- (a) Show that the rate of change of the height of water is $\frac{dh}{dt} = \frac{25}{3h^2}$. [4 marks]
- (b) Find the rate of change of the height of water after 5 seconds. [3 marks]
- (c) Given the height of the cone is 20 cm, find the time taken to fill the cone completely with water. [2 marks]

END OF QUESTION PAPER

Final Answer

SECTION A

1. (a) 2
(b) 1
2. Show, $\frac{d^2y}{dx^2} = \frac{-16t^3}{t^2 - 2^3}$
3. $\left(3, \frac{1}{2}\right)$ is maximum point.

SECTION B

1. $\frac{147}{104} + \frac{45}{104}i$

2. (a) $x = 4$
(b) $x : x < -3 \cup x : -2 < x < -1 \cup x : x > 3$
3. (a) $(f \circ g)(x) = \frac{1-5x}{x+1}$
 $(g \circ f)(x) = \frac{-6x-3}{2x+2}$
(b) $k = \frac{5}{4}$
4. (a) (i) $D_g : -1, 2 \cup 2, \infty$ (ii) $h(x) = \frac{-1}{5-4x}$
(b) Show
5. (a) 4
(b) (i) $\lim_{x \rightarrow 2^-} f(x) = -2$
 $\lim_{x \rightarrow 2^+} f(x) = 4 + B$
(ii) $A = -2, B = -6$
(iii) Show, $f(x)$ is not continuous.
6. (a) $-\frac{3}{x^{3/2}}$
(b) (i) $-\frac{2}{9}$ (ii) $-\frac{9}{2}$
7. (a) Show
(b) $\frac{1}{3} \text{ cm/s}$
(c) $t = 320s$

SET 5

TIME: 2 HOURS

SECTION A [25 marks]

This section consists of 3 questions. Answers **all** the questions given.

1. Evaluate the following limits:

(a) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 2x}.$

[3 marks]

(b) $\lim_{x \rightarrow \infty} \sqrt{\frac{5x+7}{6x-5}}.$

[3 marks]

(c) $\lim_{x \rightarrow 3^-} \frac{|4x-12|}{x-3}.$

[3 marks]

2. (a) Given $g(x) = (2-x)^3(x-1)$. Find $g'(x)$ in factored form.

[3 marks]

(b) Given $y = e^{-2x} \sin 3x$. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

Hence, show that $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = 0$

[6 marks]

3. The surface area of a balloon in the shape of a sphere is decreasing at the rate of $2 \text{ cm}^2 / \text{min}$. Find the rate at which the volume is decreasing when the radius of the balloon is 5 cm .

[7 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answers **all** the questions given.

1. Given a complex number $z = 2 + i$.

(a) Express $\bar{z} - \frac{1}{z}$ in the form $a + bi$, where a and b are real numbers.

[3 marks]

- (b) Obtain $\left| \frac{-1}{z - \frac{1}{z}} \right|$. [2 marks]
2. (a) Solve the equation $3 \log_9 x = (\log_3 x)^2$. [7 marks]
- (b) Solve $\frac{5}{x+3} - 1 \leq 0$. [4 marks]
3. Given the function $g(x) = \frac{1}{2x-5}$.
- (a) Show that $g(x)$ is a one-to-one function. Hence, find $g^{-1}(x)$. [5 marks]
- (b) Show that $g \circ g^{-1}(x) = x$. [2 marks]
4. Consider functions of $f(x) = (x-2)^2 + 1, x > 2$ and $g(x) = \ln(x+1), x > 0$.
- (a) Find $f^{-1}(x)$ and $g^{-1}(x)$. State the domain and range for each of the inverse function. [9 marks]
- (b) Obtain $(g \circ f)(x)$. Hence, evaluate $(g \circ f)(2)$. [4 marks]
5. State the conditions for continuity of $f(x)$ at $x = a$. [2 marks]
- (a) By using conditions for continuity of $f(x)$ at $x = a$, find the values of m and n such that $f(x) = \begin{cases} n - 2 \cos x, & x < 0 \\ 2 + mx^2, & 0 \leq x < 2 \\ m - x, & x \geq 2 \end{cases}$ is continuous on the interval $(-\infty, \infty)$. [7 marks]

- (b) If $m = -2$ and $n = 4$, determine whether $f(x)$ is differentiable at $x = 2$ or not.

[5 marks]

6. (a) A curve is defined by the parametric equations $x = 3t - \frac{1}{t}$ and $y = t + \frac{3}{t}$ where $t \neq 0$. Show that $\frac{dy}{dx} = \frac{t^2 - 3}{3t^2 + 1}$. Hence, find $\frac{d^2y}{dx^2}$.

[8 marks]

- (b) A curve with equation $x^2 - 3y^2 = ae^{y-2x} + by - 6$, where a and b are constants, passes through the point $(1, 2)$. Given $\frac{dy}{dx} = 1$ at $(1, 2)$, determine the values a and b .

[8 marks]

7. The function f is defined by $f(x) = \frac{\ln(x-1)}{x-1}$ for $x > 1$.

- (a) By considering the first and second derivatives of $f(x)$, show that there is only one maximum point on the graph $y = f(x)$

[7 marks]

- (b) Use the result in part 7(a) to state the exact coordinates of the maximum point.

[2 marks]

END OF QUESTION PAPER

Final Answer**SECTION A**

1. (a) 6
(b) $\sqrt{\frac{5}{6}}$
(c) -4
2. (a) $(2-x)^2(5-4x)$
(b) $\frac{dy}{dx} = e^{-2x}(-2\sin 3x + 3\cos 3x)$
 $\frac{d^2y}{dx^2} = -5e^{-2x}\sin 3x - 12e^{-2x}\cos 3x$
3. $-5 \text{ cm}^3 / \text{min}$

SECTION B

1. (a) $\frac{8}{5} - \frac{6}{5}i$
(b) 2
2. (a) $x = 1, x = 3^{\frac{3}{2}}$
(b) $(-\infty, -3) \cup [2, \infty)$
3. (a) $\frac{1+5x}{2x}$
(b) Show
4. (a) $f^{-1}(x) = \sqrt{x-1} + 2, g^{-1}(x) = e^x - 1$
 $D_{f^{-1}} = \{x > 1\}, R_{f^{-1}} = \{y > 2\}$
 $D_{g^{-1}} = \{x > 0\}, R_{g^{-1}} = \{y > 0\}$
(b) $\ln[(x-2)^2 + 2], \ln 2$
- 5 (i) $f(a)$ is defined
(ii) $\lim_{x \rightarrow a} f(x)$ exists, $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$
(iii) $f(a) = \lim_{x \rightarrow a} f(x)$
(a) $n = 4 \quad m = \frac{-4}{3}$
(b) $f(x)$ is not differentiable at $x = 2$

6. (a) Show, $\frac{d^2y}{dx^2} = \frac{20t^3}{(3t^2 + 1)^3}$

(b) $a = 5, b = -5$

7. (a) Show

(b) $\left(e + 1, \frac{1}{e}\right)$

SET 6

TIME: 2 HOURS

SECTION A [25 marks]This section consists of **3** questions. Answer **all** the questions given.

1. (a) Evaluate the following limits, if exist.

(i)
$$\lim_{x \rightarrow 2} \frac{x-2}{x^4-16}$$

[3 marks]

(ii)
$$\lim_{x \rightarrow \infty} \frac{(2-x)(x-1)}{(x-3)^2}$$

[3 marks]

- (b) Consider a function
- $f(x) = \frac{1}{2-\sqrt{x}}$
- . Find
- $\lim_{x \rightarrow \infty} f(x)$
- and state the equation of horizontal asymptotes for
- $f(x)$
- .

[3 marks]

2. (a) Consider the parametric equations,
- $x = t^2$
- , and
- $y = t^3 - 3t$
- .

Evaluate $\frac{dy}{dx}$ when $t = \sqrt{3}$.

[5 marks]

- (b) Given
- $y = 3^{2x-5}$
- . Find
- $\frac{d^2y}{dx^2}$
- when
- $x = 3$
- . Give the answer in the form of logarithm.

[4 marks]

3. The function
- $f(x) = x^3 - 6x^2 + 9x - 3$
- is defined on the interval
- $[0, 5]$
- . Find the critical points of
- $f(x)$
- on this interval and determine whether the critical points are local minimum or maximum.

[7 marks]

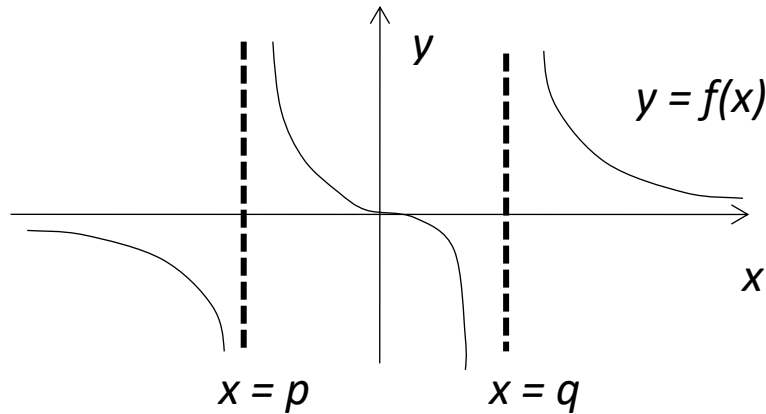
SECTION B [75 marks]This section consists of **7** questions. Answers **all** the questions given.

1. Solve for
- p
- and
- q
- where
- $p \neq q$
- , such that
- $\frac{(p+qi)}{3i} = (3 + \sqrt{-16} - i^3)$
- .

[5 marks]

2. (a) Solve the equation $\log_2(x) - \log_4(3x+4) = 0$. [4 marks]
- (b) Find the solution set for the inequality $\frac{-4-x}{x-3} \geq x+4$, $x \neq 3$. [7 marks]
3. Given $f(x) = \frac{x^2+1}{5}$, $x \geq 0$.
- (a) Determine $f^{-1}(x)$. Hence, if $f(g(x)) = \frac{1}{5}(e^{2(3x-1)} + 1)$, show that $g(x) = e^{3x-1}$. [5 marks]
- (b) Evaluate $g(f(2))$ correct to three decimal places. [2 marks]
4. Given $f(x) = e^x$ and $g(x) = |x-3|$.
- (a) Show that $(f \circ g)(x) = \begin{cases} e^{x-3} & , x \geq 3 \\ e^{-(x-3)} & , x < 3 \end{cases}$. [3 marks]
- (b) Determine $(f \circ g)^{-1}(x)$, for $x \geq 3$. [4 marks]
- (c) Find the function $h(x)$ for $x > \frac{1}{3}$, given that $(h \circ f)(x) = \frac{2e^x}{1-3e^x}$. Hence, show that $h(x)$ is a one to one function. [6 marks]
5. (a) Given $f(x) = \frac{2|x|}{x} + 5x$. Compute $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^-} f(x)$. [4 marks]
- (b) The continuous g is defined by $g(x) = \begin{cases} \sqrt{5-x} & , x < a, \\ 3x-1 & , x \geq a \end{cases}$. Find the value of a . [6 marks]

- (c) In the graph below, let $f(x) = \frac{2x}{x^2 - 4}$.



- (i) From the graph, $x = p$ and $x = q$ are the vertical asymptotes. Find the values p and q .
 (ii) Determine the horizontal asymptote.

[4 marks]

6. By writing $\tan x$ in terms of $\sin x$ and $\cos x$, show that $\frac{d}{dx}(\tan x) = \sec^2 x$.

[3 marks]

- (a) If $y = \tan x$, find $\frac{d^2 y}{dx^2}$ in terms of y . Hence, determine the range of value of x such that $\frac{d^2 y}{dx^2} > 0$ for $0 < x < \pi$.

[7 marks]

- (b) If $y = \tan(x + y)$, find $\frac{dy}{dx}$ in terms of x and y . Hence show that $\frac{dy}{dx} = -\operatorname{cosec}^2 2\alpha$ when $x = y = \alpha$.

[6 marks]

7. (a) Use the derivative to find the maximum area of a rectangle that can be inscribed in a semicircle of radius 10 cm.

[5 marks]

- (b) The side of a cube of length x cm is increasing at the rate of 2.5 cm/second. How fast is the rate of volume of the cube increasing when the side is 8 cm long?

[4 marks]

END OF QUESTION PAPER

Final Answer**SECTION A**

1. (a) (i) $\frac{1}{32}$ (ii) -1
(b) $\lim_{x \rightarrow \infty} f(x) = 0$, $y=0$ is H.A
2. (a) $\sqrt{3}$
(b) $12(\ln 3)^2$
3. Critical points $(1,1)$ and $(3,-3)$
Maximum point : $(1,1)$; Minimum point : $(3,-3)$

SECTION B

1. $p = -15, q = 9$
2. (a) $x = 4$
(b) $\{x : x \leq -4 \cup 2 \leq x < 3\}$
3. (a) $\sqrt{5x-1}$, show
(b) 7.389
4. (a) Show
(b) $\ln x + 3$
(c) $\frac{2x}{1-3x}$, $h(x)$ is one to one function
5. (a) $2, -2$
(b) $a = 1$
(c) (i) $p = -2, q = 2$ (ii) $y = 0$
6. (a) $2(1+y^2)y$, $0 < x < \frac{\pi}{2}$
(b) $\frac{\sec^2(x+y)}{1-\sec^2(x+y)}$
7. (a) 100 cm^2
(b) $480 \text{ cm}^3 / \text{second}$

SET 7

TIME: 2 HOURS

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** the questions given.

1. Evaluate the following (if exist):

(a) $\lim_{x \rightarrow 2} \frac{x^2 + 4x - 12}{|x - 2|}$ [5 marks]

(b) $\lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{1 - x}$ [4 marks]

2. Find the derivative of the following functions:

(a) $f(x) = \cot \sqrt{4x^2 + 1}$. [5 marks]

(b) $f(x) = e^{2x} \ln(3x + 4)$. [4 marks]

3. Given $f(x) = x^2(x - 2)^2$. Show that $f(x)$ has two points of inflection and state the coordinates. [7 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answers **all** the questions given.

1. Given $z_1 = 3 - 3i$. Write $\overline{z_1}$ in polar form. [5 marks]

2. (a) Evaluate the solution of $4^{y-2} = \frac{1}{3^{-y}}$ up to three decimal places. [5 marks]

(b) Solve the inequality $\left| \frac{x-1}{x+3} \right| > 2$. [6 marks]

3. Let $f(x) = \ln(3x+2)$ and $g(x) = e^{-x} + 2$ be two functions.
Evaluate $(g \circ f)^{-1}(3)$. [7 marks]
4. Given a function $f(x) = \sqrt{3-2x}$.
- (a) Show that f is a one function. [2 marks]
- (b) Find the domain and range of f . [3 marks]
- (c) Determine the inverse function of f and state its domain and range. [4 marks]
- (d) Sketch the graphs of f and f^{-1} on the same axis. [3 marks]
5. (a) Evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2+3x}}{5x+1}$ [4 marks]
- (b) Given
$$f(x) = \begin{cases} 5 - px, & -2 < x \leq -1 \\ x^2 + px + q, & -1 < x \leq 2 \\ \frac{x^2 - 4}{x - 2}, & x > 2 \end{cases}$$
- (i) Find the values of p and q if $f(x)$ is continuous for all values of x . [6 marks]
- (ii) Sketch the graph of $f(x)$ using the values p and q obtained in part (i). [4 marks]
6. (a) Given $y = \frac{\sin x}{1 + \cos x}$. Find $\frac{dy}{dx}$ in terms of x . [3 marks]
- (b) A curve is given by the parametric equations
$$x = t - \frac{1}{t}, \quad y = t + \frac{1}{t}.$$
- (i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of t . [7 marks]

(ii) Obtain the coordinates of the stationary points of the curve and determine the nature of the points.

[6 marks]

7. (a) Given $f'(x) = \frac{-2x^2 + 4x + 6}{(x^2 - x + 4)^2}$, the critical points are $\left(-1, \frac{1}{3}\right)$ and $\left(3, \frac{7}{5}\right)$. Determine the intervals where f is increasing and f is decreasing.

[3 marks]

- (b) Water is running at steady rate of $36\pi \text{ cm}^3 \text{ s}^{-1}$ into a right inverted circular cone with a semi-vertical angle of 45° .

(i) Find the rate of increasing in water depth when the water level is 3 cm.

[3 marks]

(ii) Find the time taken when the depth of the water is 18 cm.

Hint :

$$\frac{dV}{dt} = \frac{V}{t}$$

[3 marks]

END OF QUESTION PAPER

Final Answer

SECTION A

1. (a) Limit does not exist

(b) $\frac{1}{2}$

2. (a) $-\frac{4x}{\sqrt{4x^2 + 1}} \operatorname{cosec}^2(4x^2 + 1)^{\frac{1}{2}}$

(b) $(e^{2x}) \left[\frac{3}{3x + 4} + 2 \ln(3x + 4) \right]$

3. (0.423,0.445) and (1.577,0.445)

SECTION B

1. $3\sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$
2. (a) 9.638
(b) $\left\{x: -7 < x < -\frac{5}{3}, x \neq -3\right\}$
3. $-\frac{1}{3}$
4. (b) $D_f = \left(-\infty, \frac{3}{2}\right], R_f = [0, \infty)$
(c) $f^{-1}(x) = \frac{3-x^2}{2}, D_{f^{-1}(x)} = [0, \infty), R_{f^{-1}(x)} = \left(-\infty, \frac{3}{2}\right]$
(d) graph
5. (a) $-\frac{\sqrt{2}}{5}$
(b) (i) $p = -1, q = 2$
(ii) graph
6. (a) $(1 + \cos x)^{-1}$
(b) (i) $\frac{dy}{dx} = \frac{t^2 - 1}{t^2 + 1}, \frac{d^2y}{dx^2} = \frac{4t^3}{(t^2 + 1)^3}$
(ii) The stationary points : **(0, 2)** and **(0, -2)**
(0, 2) is a minimum point & (0, -2) is a maximum point.
7. (a) f is increasing in the interval $(-1, 3)$.
 f is decreasing in the interval $(-\infty, -1)$ and $(3, \infty)$.
(b) (i) 4cms^{-1}
(ii) 54s

SET 8

TIME: 2 HOURS

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** the questions given.

1. Find the limit of the following, if it exists.

(a) $\lim_{x \rightarrow -3} \frac{x+3}{x^3+27}$

[3 marks]

(b) $\lim_{x \rightarrow -\infty} \frac{2x-1}{\sqrt{x^2-9}}$

[3 marks]

(c) $\lim_{x \rightarrow 4} \frac{x^2-3x-4}{\sqrt{x}-2}$

[3 marks]

2. (a) Use the first principle to find the derivative of $g(x) = \sqrt{1-x}$.

[4 marks]

- (b) Given that $e^y + xy + \ln(1+2x) = 1$, $x \geq 0$.

Find the value of $\frac{d^2y}{dx^2}$ at the point (0,0).

[5 marks]

3. The curve $y = ax^3 + bx^2 + cx + d$ where a , b , c and d are constant, passes through the point $(0, 7)$, has a relative minimum point at $(1, 0)$ and a point of inflection at $x = -\frac{1}{2}$.

- (a) Find the values of a , b , c and d .

[3 marks]

- (b) Find all the critical points of y .

[2 marks]

- (c) Determine the nature of each critical point.

[2 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answers **all** the questions given.

1. Let $z = a + bi$ be a nonzero complex number. Find the value of a and b if
- $$z(2-i) = (\bar{z}+1)(1+i).$$

[5 marks]

2. (a) Find the solution set for the inequality

$$2 - \left(\frac{x+2}{x-4} \right) < 5.$$

[5 marks]

- (b) Solve the equation $3^x + 3^{(3-x)} = 12$.

[6 marks]

3. Sketch the graph of $f(x) = \frac{1}{2} \tan(x - \pi)$ for $0 \leq x \leq 2\pi$. Hence, state its domain and range.

[5 marks]

4. (a) Determine whether $f(x) = \frac{1}{x-4}$ and $g(x) = \frac{4x+1}{x}$ are inverse function of each other by computing their composite functions.

[5 marks]

- (b) Given $f(x) = \ln(1-3x)$.

(i) Determine the domain and range of $f(x)$. Then sketch the graph of

$$f(x).$$

[5 marks]

(ii) Find $f^{-1}(x)$, if it exists. Hence, state the domain and range of

$$f^{-1}(x).$$

[5 marks]

5.

$$\text{Given that } f(x) = \begin{cases} 1 + e^x, & x \leq 0 \\ \frac{x+6}{3-x}, & 0 < x \leq 4 \\ C, & x > 4 \end{cases}$$

where C is a constant.

(a) Determine whether $f(x)$ is continuous at $x = 0$.

[5 marks]

(b) Given that $f(x)$ is discontinuous at $x = 4$, determine the values of C .

[3 marks]

(c) Find the vertical asymptote of $f(x)$.

[6 marks]

6.

Consider the parametric equations of the curve

$$x = \cos^3 \theta \quad \text{and} \quad y = \sin^3 \theta, \quad 0 < \theta \leq 2\pi.$$

(a) Find $\frac{dy}{dx}$ and express your answer in terms of θ .

[4 marks]

(b) Find the values of $\frac{dy}{dx}$ if $x = \frac{\sqrt{2}}{4}$.

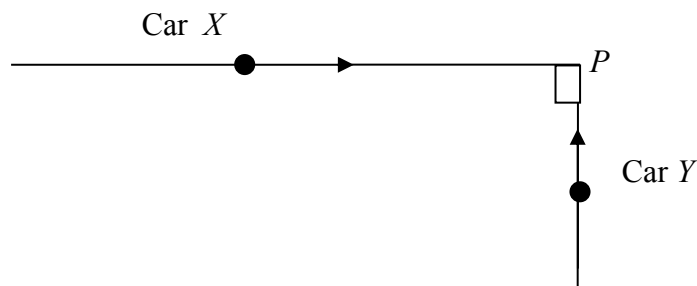
[6 marks]

(c) Show that $\frac{d^2y}{dx^2} = \frac{1}{3\cos^4 \theta \sin \theta}$. Hence, calculate $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{3}$.

[5 marks]

7.

Car X is travelling east at a speed of 80 km/h and car Y is travelling north at 100 km/h as shown in the diagram below. Obtain an equation that describes the rate of change of the distance between the two cars. Hence, evaluate the rate of change of the distance between the two cars when car X is 0.15 km and car Y is 0.08 km from P .



[9 marks]

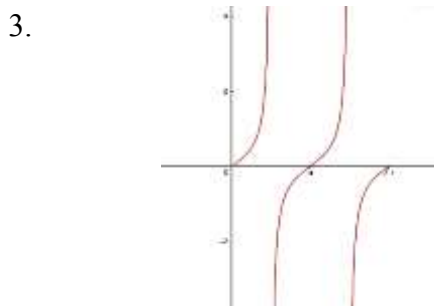
END OF QUESTION PAPER

Final Answer**SECTION A**

1. (a) $\frac{1}{27}$
 (b) -2
 (c) 20
2. (a) $\frac{-1}{2\sqrt{1-x}}$
 (b) 4
3. (a) $a = 2, b = 3, c = -12, d = 7$
 (b) $(1, 0), (-2, 27)$
 (c) $(-2, 27)$ is a relative maximum point
 $(1, 0)$ is a relative minimum point

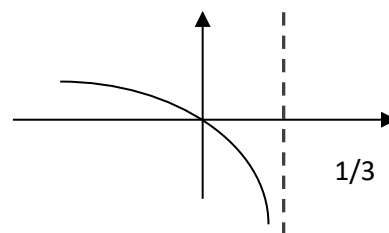
SECTION B

1. $a = 1, b = 1$
2. (a) $\left(-\infty, \frac{5}{2}\right) \cup (4, \infty)$
 (b) $x = 2$ or $x = 1$



$$D_f = \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3}{2}\pi\right) \cup \left(\frac{3}{2}\pi, 2\pi\right] \quad R_f = (-\infty, \infty)$$

4. (a) f & g are inverse function
 (b) (i) $D_f = \left(-\infty, \frac{1}{3}\right), R_f = (-\infty, \infty)$



- (b) (ii) $f^{-1}(x) = \frac{1-e^x}{3}$, $D_f = (-\infty, \infty)$, $R_f = \left(-\infty, \frac{1}{3}\right)$
5. (a) $f(x)$ is continuous at $x=0$
(b) $C \neq -10$
(c) $x = 3$
6. (a) $\frac{dy}{dx} = -\tan \theta$
(b) $-1, 1$
(c) 6.158
7. $\frac{dz}{dt} = \frac{x \frac{dx}{dt} + y \frac{dy}{dt}}{(x^2 + y^2)^{\frac{1}{2}}}$, 118 km/h